Manan Jain

Hyderabad | +91 82950 53791| mananjen@gmail.com | github.com/mananjen
linkedin.com/in/mananjen/ | mananjen.github.io/portfolio-website

SUMMARY

ML/NLP Engineer with **4 years** Software Engineering experience (Lowe's) and hands-on work building ML systems. Recent work includes **training-free multilingual image–text retrieval (Relative Representations)** using pretrained encoders (**Qwen3-Embedding-8B** for text, **c-radio-v4-h** for vision), **Hinglish sarcasm detection** (manuscript **under review**), and **RAG/LLM** Q&A over multi-TB esports data (AWS Bedrock). Tech: Python, PyTorch, TensorFlow, SQL, Transformers, AWS/GCP, React/Node.

SKILLS

Programming: Python, SQL, JavaScript, C++, HTML/CSS
ML / AI: PyTorch, TensorFlow, Transformers (Hugging Face), LLMs, RAG, prompt engineering, fine-tuning, multimodal retrieval
Data: Pandas, NumPy, scikit-learn, dataset curation, preprocessing, mining
Backend / Web: React, Node.js, REST APIs, UI component systems
Cloud & Tools: AWS (Bedrock), GCP, Git/GitHub, Jira, Confluence, Agile/Scrum

EXPERIENCE

ML Engineer, Velocitor Solutions, Hyderabad, India — July 2026 – Present

- Developing a **LangChain**-based AI agent for querying real-time vehicle-fleet telemetry data from a production lakehouse built on **Apache Spark**, **Iceberg, Kafka,** and **Kubernetes**, with exposure to LLM-driven incident triage and root-cause analysis.

Researcher (Graduate Research Assistant), UIC NLP Lab, Chicago, IL — May 2025 – July 2026

- Delivered a **training-free / non-parametric multilingual image-text retrieval** pipeline that achieves usable retrieval performance **(30% R@1)** without end-to-end model training, leveraging pretrained **unimodal vision + multilingual text** encoders and an external anchor set **(CC12M)**; **manuscript in preparation for ACL Rolling Review.** [*See Projects*]
- Led **Hinglish Sarcasm Detection**: built end-to-end pipeline, ran ablations/error analysis; **manuscript submitted to ACL Rolling Review—under review**. [*See Projects*]

Software Engineer, Lowe's, Bengaluru, India — July 2019 – August 2023

- Improved in-store associate app **usability by 30%** by optimizing UI components and flows; reduced clicks and simplified navigation.
- Revamped NPC web app components to boost customer engagement and **improve UX by 50%**, standardized UI patterns for reuse.
- Led analytics modernization: cleaned up tagging, increased code **clarity/reuse by 70%**, and migrated from Adobe DTM to Adobe Launch for more reliable tracking.

PROJECTS

Multilingual Image–Text Retrieval Without End-to-End Training, University of Illinois Chicago (PyTorch, Python, Sparse Retrieval)

- Developed a **compute-efficient alternative to end-to-end multimodal training** by converting pretrained embeddings into **anchor-relative sparse representations**, enabling usable multilingual retrieval under limited compute/data constraints.
- Used **Qwen3-Embedding-8B** for multilingual text embeddings and **c-radio-v4-h** for image embeddings; selected this pairing based on strongest empirical Recall@K performance in sweeps.
- Built scalable evaluation + orchestration over **XM3600** with **per-language Recall@K**, automated hyperparameter sweeps (*k/p*), and caching to eliminate repeated embedding computation.
- Manuscript / Github in preparation for ACL Rolling Review.

Hinglish Sarcasm Detection, University of Illinois Chicago [GitHub] (BERT, PyTorch, Data Mining, Transformers)

- Built a **BERT-based** classifier on **~20K** Hinglish/code-mixed samples; implemented preprocessing, training, and evaluation in **PyTorch/Transformers**; Manuscript submitted to ACL Rolling Review (under review).
- Ran systematic **ablations + error analysis** (code-mixing, transliteration, slang/context dependence) and documented failure modes + fixes.

Riot and Amazon Hackathon, University of Illinois Chicago [GitHub] (PyTorch, AWS, RAG)

- Bedrock-powered Q&A over ~3 TB player/match data with retrieval grounding; designed prompts/evals to reduce hallucinations.
- Built a web interface to power the solution.

Star Explorer, University of Illinois Chicago [GitHub] [Site] (C#, CAVE2 VR, Unity)

- Unity + octree rendering of millions of stars at ~70 FPS; interactive scale & constellation views.

EDUCATION

University of Illinois, Chicago | MS Computer Science | 3.66/4 — Graduated May 2025
Focus: applied NLP/ML and model-to-product execution (Data Mining, Text/NLP, Statistical NLP, AI/ML, Advanced ML, VR).

National Institute of Technology, Kurukshetra | BTech Computer Engineering | 8.11/10 — Graduated May 2019
Core: DSA, Algorithms, Operating Systems, DBMS, OOP/C++, Networks, Discrete Math & Automata, Calculus, Probability, Statistics, Software Engineering.